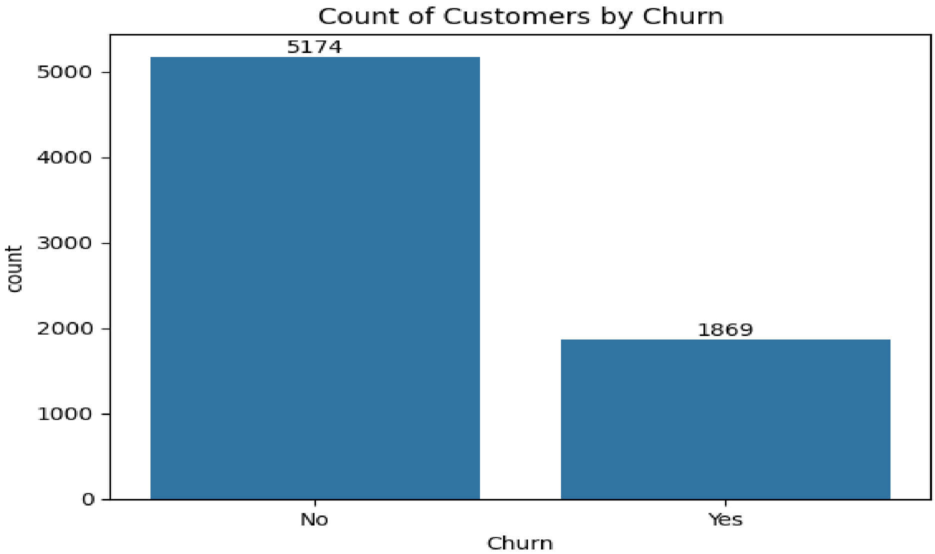
Customer Churn EDA Analysis Report Count of Customers by Churn



What this chart shows
The first thing we check is the target variable.

- No Churn = **5174 customers**
- Churn = **1869 customers**

This means

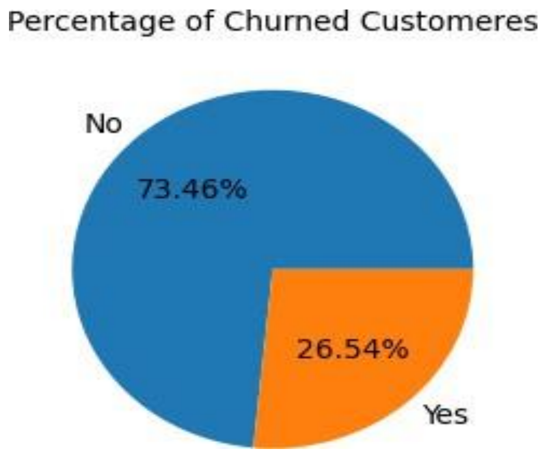
- **73.46% stayed**
- **26.54% left**

What outcome does it tell?
The dataset is **imbalanced**.
Nearly **3 out of every 4 customers stay** while only **1 out of 4 customers leave**.
Why is this important?
Machine Learning models may become biased towards predicting "No Churn."
Instead of only Accuracy, we should use

- Precision
- Recall
- F1 Score
- ROC-AUC

Business Insight
Customer churn is not extremely high, but **26.5% loss is significant**.
Reducing churn by even **5% can save millions** in customer acquisition costs.
Interview Explanation
"The target variable is moderately imbalanced with about 73% non-churn customers. Therefore, during modeling I focused on Recall and ROC-AUC instead of relying only on accuracy."

Percentage of Churn

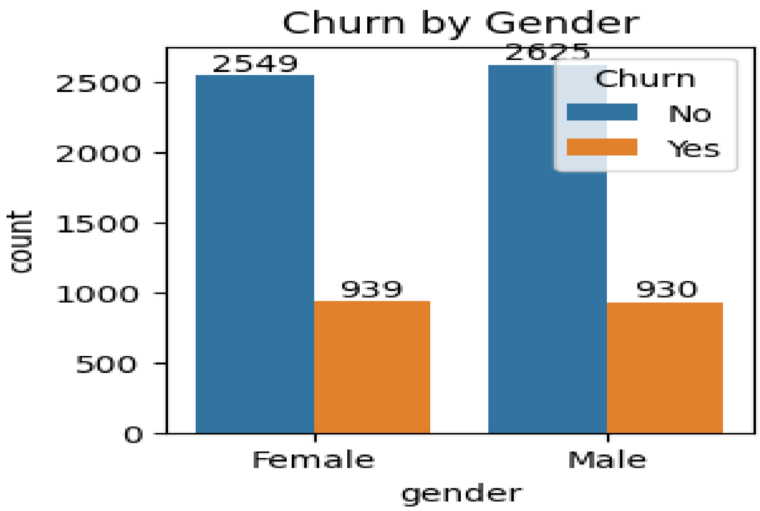


What does it show?
Same information presented visually.

- No = 73.46%
- Yes = 26.54%

Why use pie chart?
Good for executive presentations.
Managers understand percentage distribution quickly.
Insight
No severe imbalance, but enough to require attention during modeling.

Churn by Gender



Female
Stayed = 2549
Left = 939
Churn Rate
939/(2549+939)
≈ **26.9%**

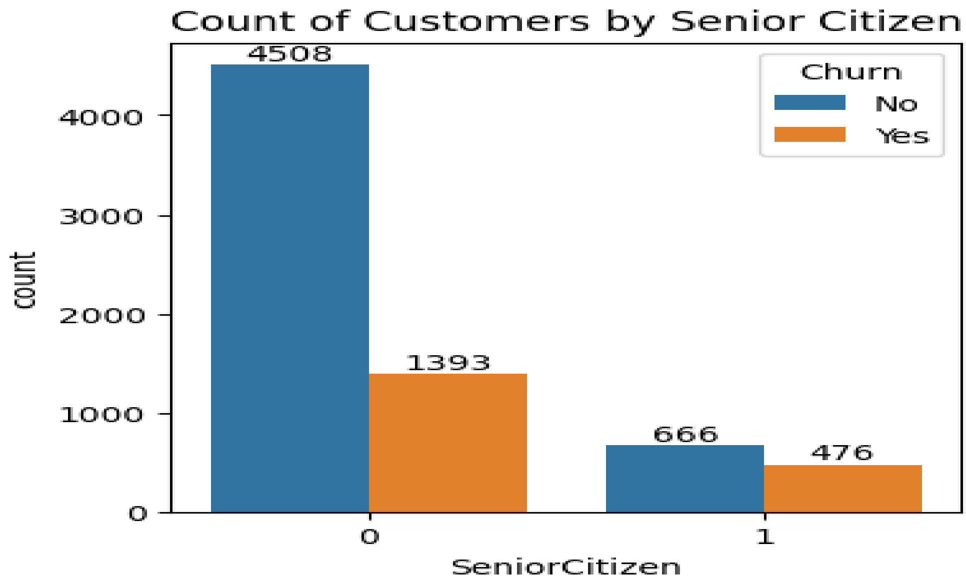
Male
Stayed = 2625
Left = 930
Churn Rate
930/(2625+930)
≈ **26.2%**

Observation
Both genders have almost identical churn rates.
Difference is less than 1%.
Business Insight
Gender has almost **no predictive power**.
The company should not create gender-based retention campaigns.
Instead focus on

- contract
- payment
- tenure

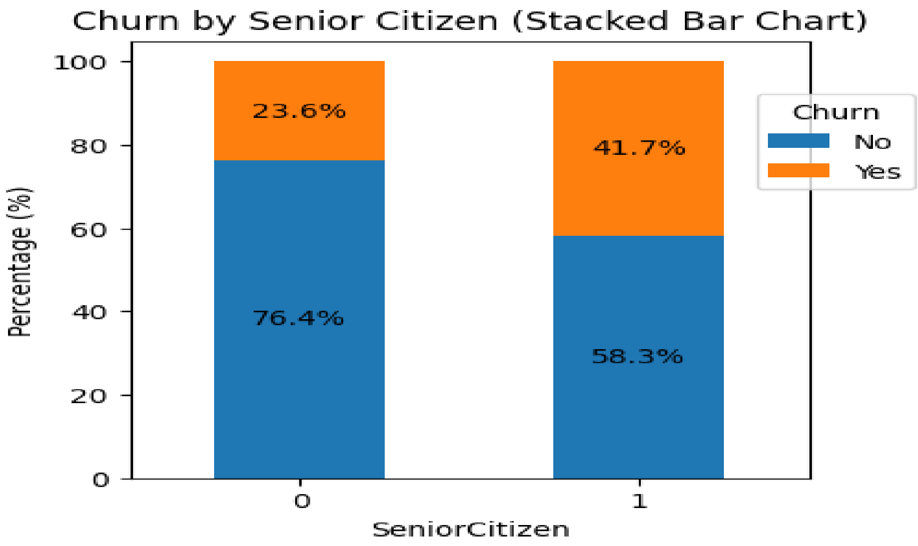
Interview Answer
"Gender doesn't appear to influence churn significantly. Male and female customers leave at nearly identical rates, indicating this feature may contribute little to prediction."

Senior Citizen



Another view of the same feature.
Non Senior
Stayed = 4508
Left = 1393
Senior
Stayed = 666
Left = 476
Observation
Senior customers are fewer in number.
But proportionally many more leave.
Analyst Insight
Absolute counts can mislead.
Always compare **percentages** instead of raw counts.
Interviewers love hearing this.

Senior Citizen Percentage



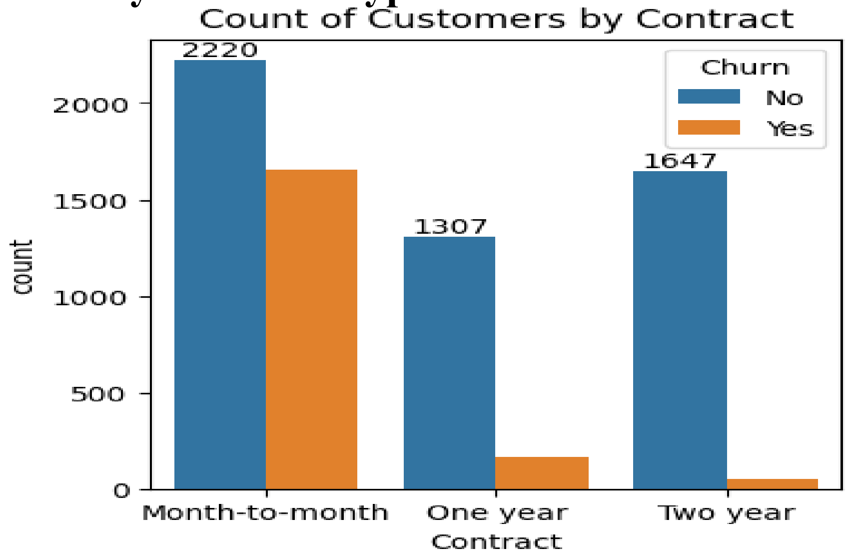
This chart is much more informative.
Non Senior
Stayed = 76.4%
Left = 23.6%
Senior
Stayed = 58.3%
Left = 41.7%

Observation
Senior citizens churn much more.
41.7%
vs
23.6%
Nearly **double**.
Business Insight
Senior customers are a vulnerable customer segment.
Possible reasons

- higher prices
- usability issues
- customer support
- fixed income

Retention strategies should target senior customers.
Interview Explanation
"Senior citizens have nearly double the churn rate compared to non-senior customers. This suggests age-related factors significantly influence customer retention."

Churn by Contract Type



Probably the strongest feature.

Month-to-Month

Stayed = 2220

Left = 1650

Huge churn.

One Year

Stayed = 1307

Left = 166

Very low churn.

Two Year

Stayed = 1647

Left = 53

Almost no churn.

Business Insight

Long-term contracts dramatically reduce churn.

This is expected because

Customers who commit longer are

- more loyal
- invested
- receive discounts

Recommendation

Convert Month-to-Month customers into

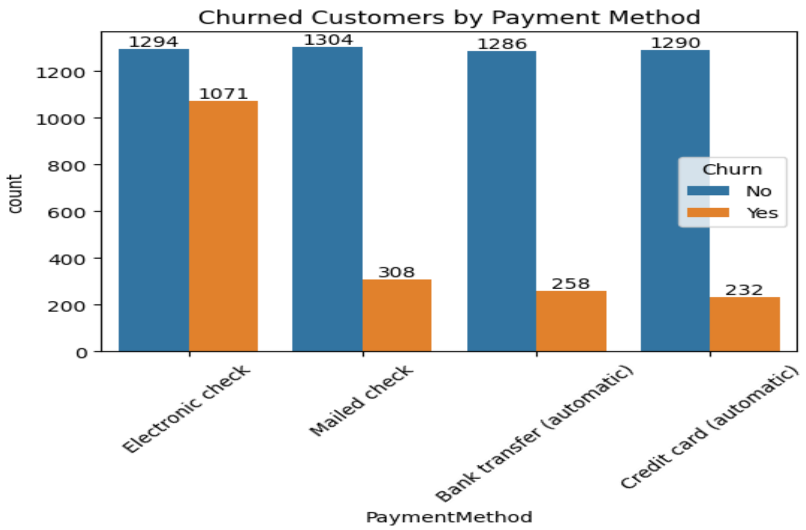
- yearly
- two-year plans

This single strategy could significantly reduce churn.

Interview Explanation

"Contract type is one of the strongest predictors. Customers on month-to-month contracts churn at much higher rates, while two-year contracts have extremely low churn."

Payment Method



Electronic Check
Stayed = 1294
Left = 1071
Huge churn

Mailed Check
Stayed = 1304
Left = 308

Bank Transfer
Stayed =1290
Left =258

Credit Card
Stayed =1290
Left =232
Observation
Electronic Check customers churn much more.
Possible Reasons
Could indicate

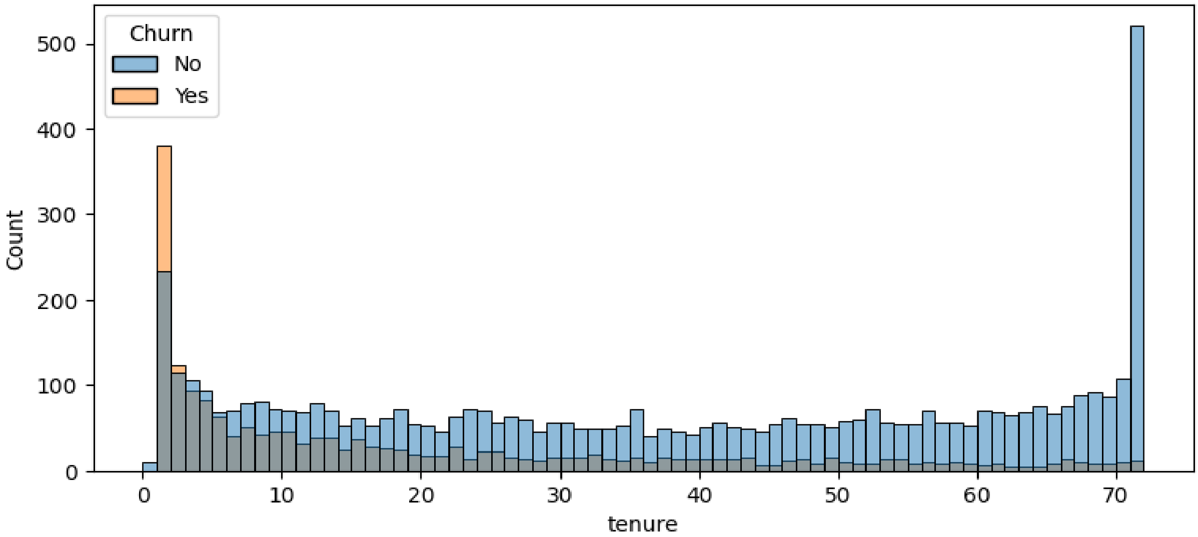
- dissatisfied customers
- temporary users
- payment inconvenience
- no auto-renewal

Business Recommendation
Encourage

- Auto payment
- Credit Card
- Bank Transfer

Offer discounts for automatic billing.
Interview Explanation
"Electronic check users show the highest churn among payment methods, indicating payment experience or customer profile differences."

Tenure Distribution



This histogram is extremely useful.
Notice
Large spike
at
0-3 months
Most churn happens here.
Another spike exists at
72 months
Very loyal customers.

Observation

New customers leave quickly.
Customers surviving beyond 2-3 years rarely churn.

Business Insight

Customer lifecycle has two phases
Early Risk
↓
Long-term Loyalty

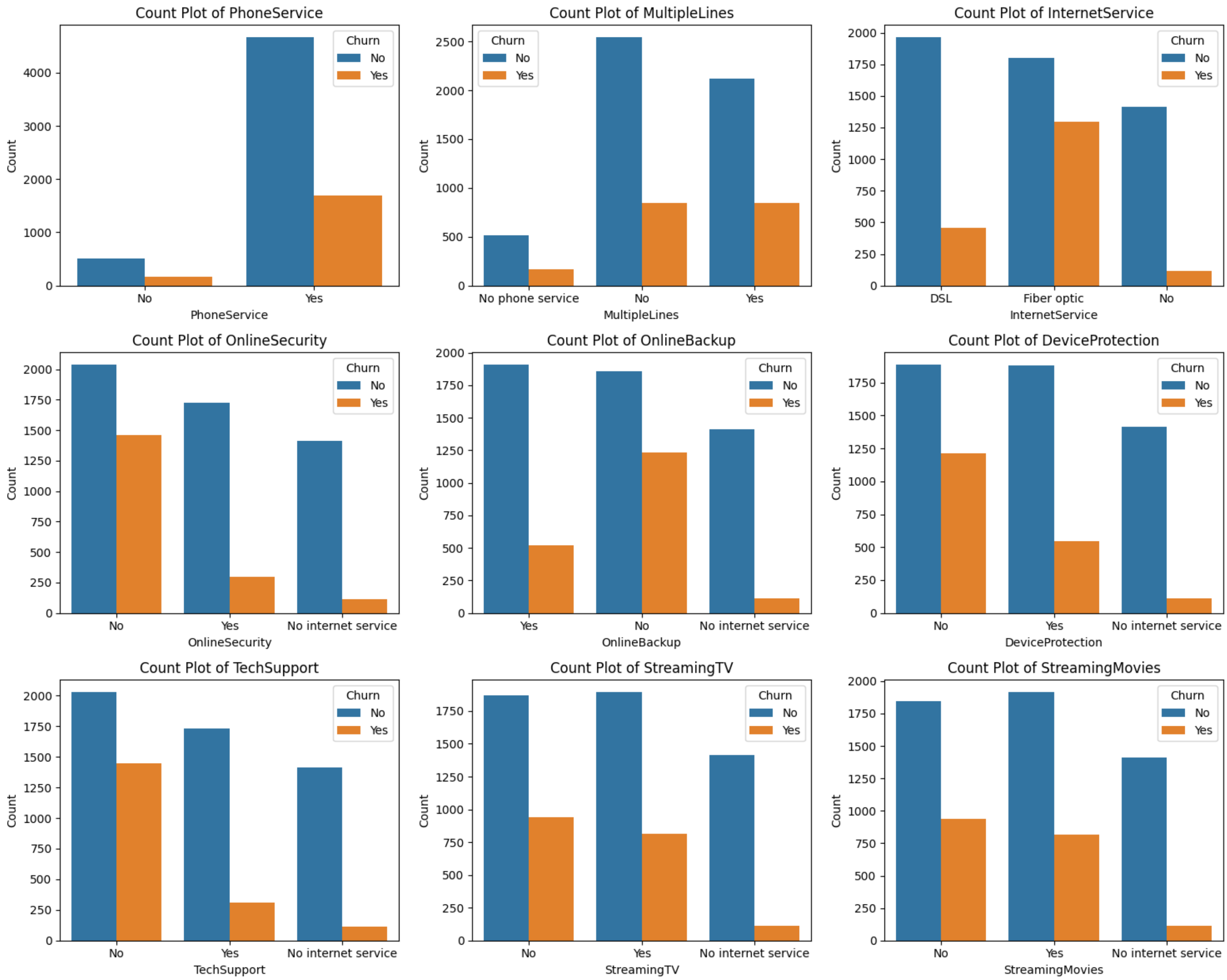
Recommendation

Focus retention campaigns within the first
90 days.

Interview Explanation

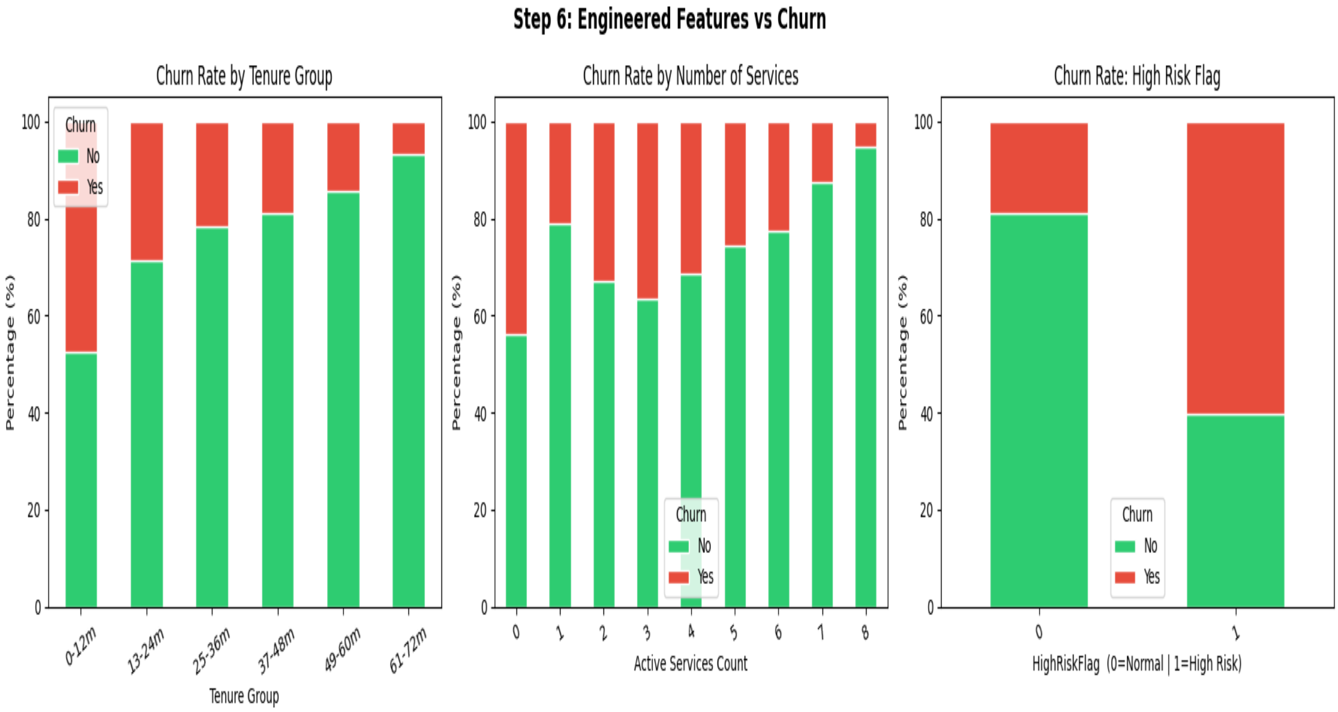
"Tenure shows that most churn occurs among newer customers, while long-tenure customers remain highly loyal."

Service Features



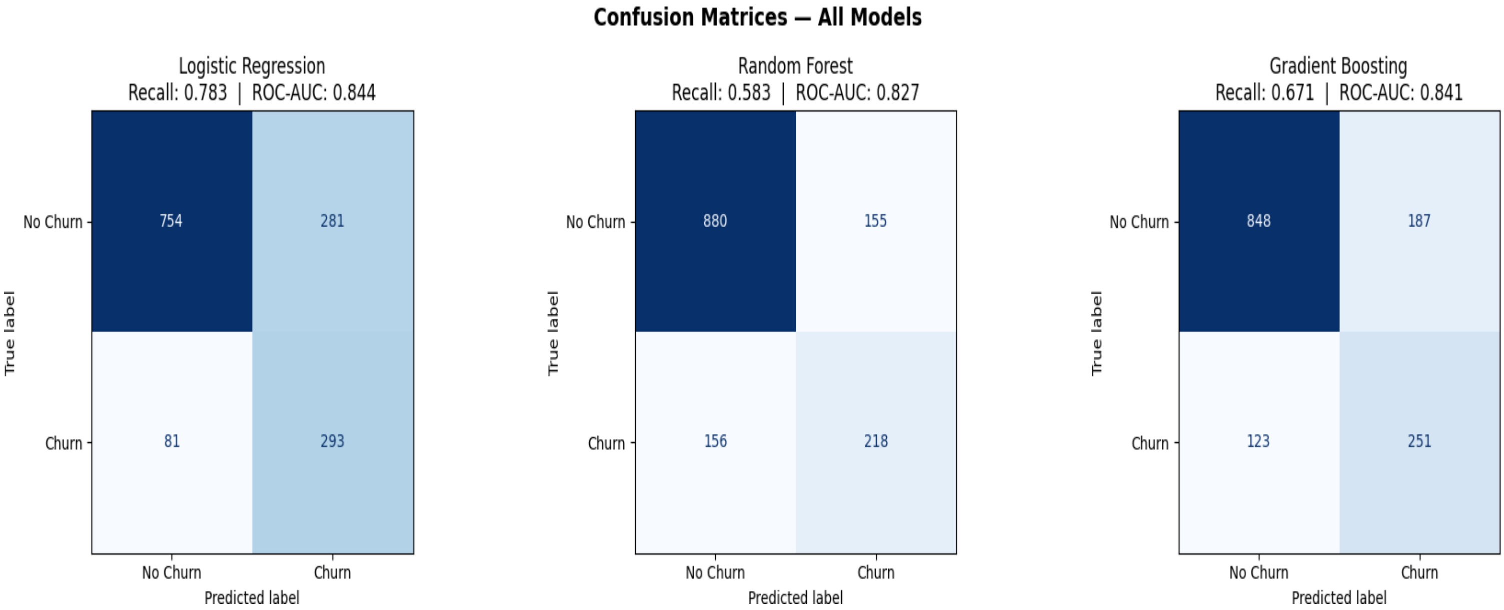
Observation: Customers without Online Security, Tech Support, Backup and Protection churn more. Insight: Additional services increase customer stickiness.

Engineered Features



Observation: More active services reduce churn; HighRiskFlag identifies high-risk customers. Insight: Feature engineering improved business understanding.

Confusion Matrices



Logistic Regression achieved the highest recall and strongest overall balance.
Interview: Explain TN, FP, FN, TP and emphasize recall for churn problems.

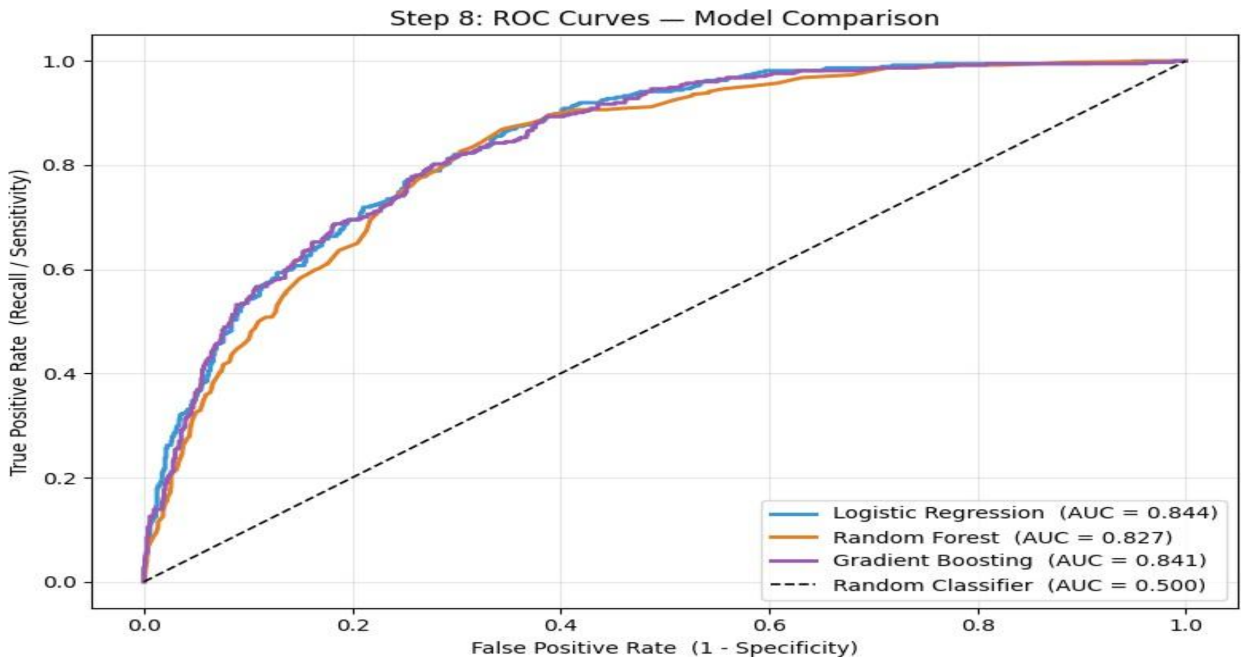
Logistic Regression
TN = 754
FP =281
FN =81
TP =293
Recall =0.783
ROC=0.844
Interpretation
Caught
293 churners
Missed
81
High recall.
Good for churn prediction.

Random Forest
Recall =0.583
Missed many churn customers.
Better specificity.
Poor recall.

Gradient Boosting
Recall =0.671
Balanced performance.

Business Insight
Missing churn customers is expensive.
Therefore Higher Recall is preferred.

ROC CURVE



Logistic Regression AUC=0.844, Gradient Boosting=0.841, Random Forest=0.827. Insight: Logistic Regression performs best despite being simpler and more interpretable.

AUC
Logistic Regression
0.844
Gradient Boosting
0.841
Random Forest
0.827
Observation
All models perform well.
Logistic Regression slightly wins.
Interesting because
A simpler model outperformed complex models.
This often happens with structured tabular datasets.
Interview Explanation
"Although ensemble methods are powerful, Logistic Regression achieved the highest ROC-AUC of 0.844, indicating that the relationships in this dataset are largely linear and well captured by a simpler interpretable model."

ROC CURVE

SUMMARY: The EDA revealed several strong drivers of customer churn. The most influential factors were contract type, tenure, payment method, senior citizen status, and the adoption of value-added services such as online security and technical support. Customers on month-to-month contracts, using electronic checks, with shorter tenure, and lacking additional services showed substantially higher churn rates. These insights suggest practical business actions, including improving onboarding during the first 90 days, encouraging migration to long-term contracts, promoting automatic payment methods, and bundling support and security services to increase customer retention. From a modeling perspective, the dataset showed moderate class imbalance (73% vs. 27%), so evaluation focused on Recall and ROC-AUC rather than accuracy alone. Logistic Regression achieved the best overall performance with an ROC-AUC of 0.844 while maintaining the highest recall, making it the most suitable model for identifying customers at risk of churning.